Supplementary material for the work Path Integral Monte Carlo on a Sphere

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Supplementary material to the article "Path Integral Monte Carlo on a Sphere".

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I. INTRODUCTION

We here report the FORTRAN code listing of the code used in the simulations developed in the publication "Path Integral Monte Carlo on a Sphere". This is shown in Appendix A. The model studied in that publication was that of a many body fluid of N particles at $\boldsymbol{r}_i$, $i = 1, 2, \ldots, N$, on a sphere. Where $\boldsymbol{r} = (\theta, \varphi)$ with $\theta \in]-\pi/2, \pi/2]$ the polar angle and $\varphi \in]-\pi, \pi]$ the azimuthal angle.

The plane waves [1] Path Integral Monte Carlo (PIMC) method used in our computer experiments is the one described in Ref. [2]. A single MC step is made of a swap of a randomly chosen pair of particles through the *Lévy construction* of a Brownian bridge between two randomly chosen timeslices on different paths (see Section V.G of Ref. [2] and references therein for the multislice move) and of a displacement of all the timeslices of a particle path chosen at random (the singleslice move). This allows the permutation sampling necessary in calculating the permanent required by the study of the Bose-Einstein statistics. The extent of the singleslice displacement is chosen at the beginning of the run through the input variable `ALPHA` or it can be adjusted during the run every `IRATIO` MC steps so to have acceptance ratios closest to 1/2 (this functionality is commented out in the present listing).

We discretize the imaginary time between 0 and `BETA` into `FTNO` timeslices of length τ =`LS`=`BETA`/`FTNO`. One usually wants to compare simulations at fixed `LS`. The multislice move requires the reconstruction of a randomly chosen number `MBM`≤`MBMM`≤`FNTO` of timeslices between a randomly chosen timeslice `CP` on a randomly chosen particle `IP` and a timeslice on a randomly chosen particle `KP`. Here `MBMM` is the maximum number of timeslices in a Brownian bridge. The singleslice move requires the displacement of all the `FTNO` timeslices of the path of the particle `IP` chosen at random. So a single MC step requires a maximum of `MBMM`×`FTNO` slice moves. In the code we wrote the FORTRAN instructions for the multislice move in lowercase and the ones for the singleslice move in uppercase. The simulation uses only the multislice move if the logical variable `IFBRIDGE` is `TRUE` and only the singleslice move if the logical variable `IFDISP` is `TRUE`. The simulation lasts for a total number of MC steps `NSTEP`.

On a sphere it is necessary to choose `DIM`= 2 with a number of particles N =`NP` of mass `FTM` with natural units $\hbar = k_B = 1$. One has to specify also the radius of the sphere `RADIUS` and the absolute temperature `TEMP`.

The pair potential between the particles `POTK` is encoded in `SUBROUTINE POT` at the end of the code listing. It includes the Coulomb potential (`COULOMB`). The ideal gas is obtained choosing the string `FREE` in the `data-gr-rs.in` simulation configuration input file. The variables `RCUT`, `SIG`, `EPSR`, `EPSA`, and `EPS` in the simulation configuration input file are used to specify other properties of a non Coulomb pair potential.

The simulation calculates both thermodynamic quantities and static structural properties. For the thermodynamic quantities we measure: the kinetic energy `KE`, the physical potential energy `V`, the geometric effective potential energy `W`, $-\sum_{i=1}^{N} \ln \sqrt{g(\boldsymbol{r}_i)}/\tau$ where g is the determinant of the sphere metric tensor, and the superfluid fraction `SF`. These quantities are stored in the output file `fort.9` every `IPRINT` MC steps. Whereas the path configurations output file `CNFILE` is stored in a *.xyz* format every `ISAVE` MC steps. The first `IEQUI` MC steps are used for the equilibration of the Markov chain. Quantities are only calculated after these equilibration steps. For the structural properties we measure the radial distribution function $g(r)$, with r the Euclidean distance, which is written in the output file `rdf.dat`. In the current implementation of the code, the $g(r)$ is only calculated during the displacement move every `ICALCG` timeslices.

The initial path configurations file is created at the beginning of the run if the variable `INIT` is set to 0, to start from a path with all particles and all their timeslices on the origin, to 1, to start from a path with the particles randomly distributed on the equator with all timeslices copied above the first one for each particle, to 2, to read it from the file `CNFILE` stored on disk, to 3, for a complete restart from the file `CNFILE` and the frozen configuration `back.dat` both stored on disk.

The logical variable `IFB` is set in the simulation configuration input file. If it is set to `TRUE` Bose-Einstein statistics is used, otherwise is used Boltzmann statistics.

The variable `SEED` is also set in the simulation configuration input file. It is the positive integer seed of the

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[1] For a coherent states PIMC see Ref. [1].

pseudo random number generator.

For the RPIMC algorithm also used in the publication for the simulations of fermions [3, 4] and anyons we modified the code presented here as described in the main text of the work.

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Appendix A: The code

This is the code used for the PIMC computer experiment. We list here the main FORTRAN code `pimc-sph-rs.f` with the included `mc-sph.par` parameters file. And the input simulation configuration file `data-gr-rs.in` that is read at the beginning of the run.

```
*****************************************************************************
*** pimc-sph-rs.f ***********************************************************
*****************************************************************************

      PROGRAM PIMC
      IMPLICIT NONE
C     ****************************************************************
C     ** MONTE CARLO SIMULATION PROGRAM FOR [PW] PATH INTEGRAL       **
C     **                                                             **
C     ** on a sphere of radius RADIUS                                **
C     **                                                             **
C     ** space dimensions:    DIM=2                                  **
C     ** number of particles: NP                                     **
C     ** number of timesteps: FTN0                                   **
C     ** units:               hbar=k_B=1                             **
C     ** BETA=1/TEMP=FTN0*LS  (LS imaginary time spacing)            **
C     **                                                             **
C     ** OBSERVABLES:                                                **
C     ** kinetic energy       KE                                     **
C     ** potential energy     V                                      **
C     ** forces virial        W                                      **
C     ** superfluid fraction  SF                                     **
C     ** radial distribution function on 'rdf.dat'                  **
C     ****************************************************************
      INCLUDE       'mc-sph.par'                        ! parameters

      INTEGER*4     SEED
      INTEGER*8     IDIM, STEP, I, J, K
      INTEGER*8     C1, CP, IP, KP, NIP, NKP, LL, OUT1, OUT2, PIP
      INTEGER*8     INIT, NSTEP, NS, IPRINT, ISAVE, IRATIO, IEQUI
      INTEGER*8     MBMM, MCM, PREV(MNP)
      INTEGER*8     CYC(MNP), NSWAP

      REAL*8        DENS, TEMP, BETA, DENSLJ
      REAL*8        DRMAX, ALPHA, CDIM
      REAL*8        RANF, DUMMY, SR9, SR3
      REAL*8        VLRC, VLRC6, VLRC12, WLRC, WLRC6, WLRC12
      REAL*8        ACM, ACATMA, ACATMAS, ACATMAB
      REAL*8        V, VNEW, VOLD, VEND, DELTV, VN, VS
      REAL*8        W, WNEW, WOLD, WEND, DELTW, WN, WS
      REAL*8        KE, KENEW, KEOLD, DELTKE, KEEND
      REAL*8        AVV, ACV, ACVSQ, FLV
      REAL*8        AVW, ACW, ACWSQ, FLW
      REAL*8        AVKE, ACKE, ACKESQ, FLKE
      REAL*8        ACT, ACTNEW, ACTOLD, DELTACT, DELTACTB
      REAL*8        RXIOLD(MDIM), RXINEW(MDIM)
      REAL*8        RXP(MDIM,0:N), RXPP(MDIM,0:N)
      REAL*8        STD, PS, KKK, VVV, WWW
      REAL*8        RATIO, RATIOS, RATIOB
      REAL*8        WIND, ACWIND, SFI, SF, IC
      REAL*8        GA

      INTEGER*8     BIN, MAXBIN, MAXBIN2, HIST(1000000), ICALCG
      REAL*8        CONST, RLOWER, RUPPER
      REAL*8        TSTEP, NIDEAL, GR

      CHARACTER     CNFILE*30, POTK*10

      LOGICAL       OVRLAP, IFB, IFDISP, IFBRIDGE

C      PI = ACOS(-1.d0)
      OPEN(UNIT = 16, FILE = 'back.dat', STATUS ='UNKNOWN')

C     ****************************************************************
C     ** READ INPUT DATA                                             **
C     ****************************************************************

      WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)')
      WRITE(*,'('' PLANE WAVES                                  ''')')
      WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM            ''')')

      OPEN (UNIT=10, FILE='data-gr-rs.in', STATUS='UNKNOWN')
      READ (10,*) I
      READ (10,*) DIM
      READ (10,*) I
      READ (10,*) SEED
      READ (10,*) I
      READ (10,*) IFB
      READ (10,*) I
      READ (10,*) NP
      READ (10,*) I
      READ (10,'(A)') POTK
      READ (10,*) I
      READ (10,*) FTN0
      READ (10,*) I
      READ (10,*) FTM
      READ (10,*) I
      READ (10,*) NSTEP
      READ (10,*) I
      READ (10,*) IPRINT
      READ (10,*) I
      READ (10,*) ISAVE
      READ (10,*) I
      READ (10,*) IEQUI
      READ (10,*) I
      READ (10,*) IRATIO
      READ (10,*) I
      READ (10,*) ICALCG
      READ (10,*) I
      READ (10,'(A)') CNFILE
      READ (10,*) I
      READ (10,*) INIT
      READ (10,*) I
      READ (10,*) RADIUS
      READ (10,*) I
      READ (10,*) TEMP
      READ (10,*) I
      READ (10,*) ALPHA
      READ (10,*) I
      READ (10,*) MBMM
      READ (10,*) I
      READ (10,*) RCUT
      READ (10,*) I
      READ (10,*) SIG
      READ (10,*) I
      READ (10,*) EPSR
      READ (10,*) I
      READ (10,*) EPSA
      READ (10,*) I
      READ (10,*) EPS
      READ (10,*) I
      READ (10,*) IFDISP
      READ (10,*) I
      READ (10,*) IFBRIDGE
      CLOSE (UNIT=10)
      WRITE(*,'('' IN DIMENSION (1,2,3,...)''I12/)') DIM
      WRITE(*,'('' ***********************************      ''/)')

      GOTO 1111                    ! comment if input from keyboard

      WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)')
      WRITE(*,'('' PLANE WAVES                                  ''')')
      WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM            ''')')

      WRITE(*,'('' ENTER THE SPATIAL DIMENSIONS                 ''')')
      READ (*,*) DIM
      WRITE(*,'('' ENTER SEED FOR RANDOM SEQUENCE               ''')')
      READ (*,*) SEED
      WRITE(*,'('' IF BOSE (T/F)                                ''')')
      READ (*,*) IFB
      WRITE(*,'('' ENTER THE NUMBER OF PARTICLES < MNP          ''')')
      READ (*,*) NP
      WRITE(*,'('' ENTER TYPE OF PAIR POTENTIAL (Many Body)     ''')')
      READ (*,'(A)') POTK
      WRITE(*,'('' ENTER THE NUMBER OF DISCRETIZATIONS < N      ''')')
      READ (*,*) FTN0
      WRITE(*,'('' ENTER THE BARE MASS                          ''')')
      READ (*,*) FTM
      WRITE(*,'('' ENTER NUMBER OF CYCLES                       ''')')
      READ (*,*) NSTEP
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN OUTPUT LINES ''')')
      READ (*,*) IPRINT
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN DATA SAVES     ''')')
      READ (*,*) ISAVE
      WRITE(*,'('' ENTER NUMBER OF STEPS FOR EQUILIBRATION      ''')')
      READ (*,*) IEQUI
      WRITE(*,'('' ENTER INTERVAL FOR UPDATE OF MAX. DISPL.     ''')')
      READ (*,*) IRATIO
      WRITE(*,'('' ENTER INTERVAL FOR RDF CALCULATION           ''')')
      READ (*,*) ICALCG
      WRITE(*,'('' ENTER THE CONFIGURATION FILE NAME            ''')')
      READ (*,'(A)') CNFILE
      WRITE(*,'('' ENTER 0 INIT Z, 1 INIT, 2 REST C, 2 REST F ''')')
      READ (*,*) INIT
      WRITE(*,'('' ENTER THE SPHERE RADIUS                      ''')')
      READ (*,*) RADIUS
      WRITE(*,'('' ENTER THE TEMPERATURE                        ''')')
      READ (*,*) TEMP
      WRITE(*,'('' ENTER MAX. DISPLACEMENT DRMAX/SIGMA          ''')')
      READ (*,*) ALPHA
      WRITE(*,'('' ENTER MAXIMUM NUMBER OF BRIDGE TIMESLICES    ''')')
      READ (*,*) MBMM
      WRITE(*,'('' ENTER THE POTENTIAL CUTOFF DISTANCE (~ 2.5)''')')
      READ (*,*) RCUT
      WRITE(*,'('' ENTER SIG                                    ''')')
      READ (*,*) SIG
      WRITE(*,'('' ENTER EPSR                                   ''')')
      READ (*,*) EPSR
      WRITE(*,'('' ENTER EPSA                                   ''')')
      READ (*,*) EPSA
      WRITE(*,'('' ENTER EPS                                    ''')')
      READ (*,*) EPS
```

```
      WRITE(*,'('' IF DISPLACEMENT MOVE (T/F)                 '')')
      READ (*,*) IFDISP
      WRITE(*,'('' IF BRIDGE MOVE (T/F)                       '')')
      READ (*,*) IFBRIDGE
      WRITE(*,'('' IN DIMENSION (1,2,3,...)''I12/)') DIM
      WRITE(*,'('' ***********************************      ''/)')

 1111 CONTINUE

      IF (MBMM .GT. FTN0) THEN
         WRITE(*,*) "number of timeslices in bridge > ",FTN0
         STOP
      ENDIF

      IF (DIM .NE. 2) THEN
         WRITE(*,*) "DIM not equal to 2"
         STOP
      ENDIF

C     ** INITIALIZE RANDOM NUMBER GENERATOR **

      CALL SRAND ( SEED )
      IF (SEED.EQ.1) THEN
         CALL SRAND ( 0 )
      ENDIF

C     ** DENSITY **

      DENS = NP/4.d0/PI/RADIUS**2

C     ** INVERSE TEMPERATURE BETA **

      BETA = (1.d0/TEMP)

C     ** IMAGINARY TIMESTEP **

      LS  = BETA/FTN0      ! time step
      STD = (LS/FTM)**0.5  ! standatd deviation of free-particle

C     ** MOMENT OF INERTIA OF A UNITARY SPHERICAL SHELL **

      IC = 2*(NP*FTM)/3

C    ** WRITE INPUT DATA **

      WRITE(*,'('' SEED                       '',I10   )') SEED
      WRITE(*,'('' POTENTIAL                  '',A     )') POTK
      WRITE(*,'('' DIMENSIONS                 '',I10   )') DIM
      WRITE(*,'('' NUMBER OF PARTICLES        '',I10   )') NP
      WRITE(*,'('' NUMBER OF CYCLES           '',I10   )') NSTEP
      WRITE(*,'('' NUMBER OF EQUIL. STEPS     '',I10   )') IEQUI
      WRITE(*,'('' OUTPUT FREQUENCY           '',I10   )') IPRINT
      WRITE(*,'('' SAVE FREQUENCY             '',I10   )') ISAVE
      WRITE(*,'('' RATIO UPDATE FREQUENCY     '',I10   )') IRATIO
      WRITE(*,'('' CONFIGURATION FILE  NAME   '',A     )') CNFILE
      WRITE(*,'('' TEMPERATURE                '',E10.4 )') TEMP
      WRITE(*,'('' DENSITY                    '',E10.4 )') DENS
      WRITE(*,'('' PARTICLE MASS              '',E10.4 )') FTM
      WRITE(*,'('' # TIME SLICES              '',I10   )') FTN0
      WRITE(*,'('' TIME STEP                  '',E10.4 )') LS

C    ** CONVERT INPUT DATA TO PROGRAM UNITS **

C       SIGMA  = ( DBLE ( NP ) / DENS ) ** ( 1.0 / DIM )
      SIGMA   = 2*PI
      SIGM(1) = PI
      SIGM(2) = 2*PI
      DRMAX   = ALPHA * SIGMA

      IF (POTK .EQ. 'LJ') THEN
         DENSLJ = DENS * SIG ** DBLE( DIM )
C         RCUT = SIGMA/2.d0/SIG
      ENDIF

      MAXBIN  = INT ( 2*RADIUS/DELR )     ! Euclidean
C      MAXBIN  = INT (PI*RADIUS/DELR )     ! Geodesic
      MAXBIN2 = INT ( MAXBIN/2.0 )
      DO I = 1, MAXBIN
         HIST(I) = 0
      ENDDO

      IF (DIM .eq. 3) CONST = 4.d0 * PI * DENS / 3.d0
      IF (DIM .eq. 2) CONST = PI * DENS
      IF (DIM .eq. 1) CONST = 2.d0 * DENS

C     ** INITIALIZE CONFIGURATION **

      DO I = 1, NP
         NEXT(I) = I
         LEXT(I) = I
         DO J = 1, FTN0
            MEXT(J,I)  = I
         ENDDO
      ENDDO

C     ** ZERO ACCUMULATORS **
      ACV     = 0.0
      ACVSQ   = 0.0
      FLV     = 0.0
      ACW     = 0.0
      ACWSQ   = 0.0
      FLW     = 0.0
      ACKE    = 0.0
      ACKESQ  = 0.0
      FLKE    = 0.0
      ACWIND  = 0.0
      ACM     = 0.0
      ACATMA  = 0.0
      ACATMAB = 0.0
      ACATMAS = 0.0

      IF ( INIT .EQ. 0 ) THEN
         PRINT*,"PATHS INITIALIZATION ..............."
         RX=0.                  ! all paths zeroed
         NS = NSTEP
      ELSEIF ( INIT .EQ. 1) THEN
         PRINT*,"PATHS INITIALIZATION ..............."
C     ** READ INITIAL CONFIGURATION **
         CALL INITCNSPH ( CNFILE ) ! random configuration on equator
         CALL READCN ( CNFILE )
         NS = NSTEP
      ELSEIF ( INIT .EQ. 2) THEN
         PRINT*,"RESTART FROM CONFIG  ..............."
C     ** READ INITIAL CONFIGURATION **
         CALL READCN ( CNFILE )
         NS = NSTEP
      ELSEIF ( INIT .EQ. 3) THEN
         PRINT*,"RESTART FULL         ..............."
         CALL READCN ( CNFILE )
         READ(16,*)(HIST(I),I=1,MAXBIN),
     &          ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWIND,ACM,NS
         NS=NS+NSTEP
      ELSE
         WRITE (*,*) 'INIT different from 0,1,2,3'
         STOP
      ENDIF

C     ** CALCULATE LONG RANGE CORRECTIONS    **
C     ** SPECIFIC TO THE LENNARD JONES FLUID **
      IF (DIM .EQ. 1) CDIM = 1
      IF (DIM .EQ. 2) CDIM = PI
      IF (DIM .EQ. 3) CDIM = 2*PI

      SR3 = - RCUT ** (- 6. + DIM)/(-6. + DIM)
      SR9 = - RCUT ** (- 12. + DIM)/(-12. + DIM)

      VLRC12 =   4 * EPS * CDIM * DENSLJ * NP * SR9
      VLRC6  = - 4 * EPS * CDIM * DENSLJ * NP * SR3
      VLRC   =   VLRC12 + VLRC6
      WLRC12 =   4.0  * VLRC12
      WLRC6  =   2.0  * VLRC6
      WLRC   =   WLRC12 + WLRC6

C     ** WRITE OUT SOME USEFUL INFORMATION **

      WRITE(*,'('' SIGMA                      '',E10.4)') SIGMA
      WRITE(*,'('' MAXIMUM DISPLACEMENT       '',E10.4)') DRMAX

C    ** CALCULATE INITIAL ENERGY AND CHECK FOR OVERLAPS **

      CALL SUMUP (POTK, OVRLAP, KE, V, W)

      IF (POTK .EQ. 'LJ') THEN
         VS = ( V + VLRC )
         WS = ( W + WLRC )
      ELSE
         VS = V
         WS = W
      ENDIF

      WRITE(*,'('' INITIAL V                  '',E10.4)') VS
      WRITE(*,'('' INITIAL W                  '',E10.4)') WS
      WRITE(*,'('' INITIAL KE                 '',E10.4)') KE

      WRITE(*,'(//'' START OF MARKOV CHAIN                ''//)')
      WRITE(*,'(''  NMOVE     RATIO       ACTION          ''/)')

C     ****************************************************************
C     ** LOOPS OVER ALL CYCLES AND ALL TIME SLICES                   **
C     ****************************************************************

      DO 100 STEP = NS-NSTEP+1, NS

         CP = INT(FTN0*RANF(DUMMY))+1    ! select a random timeslice
         MBM = INT((MBMM-1)*RANF(DUMMY))+2  ! # timeslices in bridge
         IP = INT(NP*RANF(DUMMY))+1      ! select a particle
         KP = INT(NP*RANF(DUMMY))+1      ! select another particle
         IF (.NOT.IFB) KP = IP           ! for boltzmann statistics
         IF (IFDISP) GOTO 77   ! uncomment if only displacement move

C     ****************************************************************
C     ** BRIDGE&SWAP MOVE (BOSE STATISTICS)                          **
C     ****************************************************************

C     direct mapping from sphere rx to plane rp
         call mappingd()
         call equiv(rpn,rp)

         mcm = cp+mbm-floor((cp+mbm-.1)/ftn0)*ftn0

         nip=next(ip)
         nkp=next(kp)

         vold=0.d0
         wold=0.d0
         if(cp+mbm.le.ftn0)then
            do i=cp+1,mcm-1
               call pppnergy ( potk, rx(:,i,ip), ip,
     :              rx(:,i,kp), kp, i,
     :              vvv, www )
               vold=vold+vvv
               wold=wold+www
            enddo
         else
            do i=cp+1,ftn0
               call pppnergy ( potk, rx(:,i,ip), ip,
     :              rx(:,i,kp), kp, i,
     :              vvv, www )
               vold=vold+vvv
               wold=wold+www
            enddo
            do i=1,mcm-1
```

```
                 call pppnergy ( potk, rx(:,i,nip), nip,
     :                rx(:,i,nkp), nkp, i,
     :                vvv, www )
                 vold=vold+vvv
                 wold=wold+www
              enddo
           endif

           keold=0.d0
           do k=1,np
              if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
                 do i=1,ftn0
                    call kknergy ( rx(:,i,k), k, i,
     :                   kkk )
                    keold=keold+kkk
                 enddo
              endif
           enddo

c     bridge move
           if(cp+mbm.le.ftn0)then
              call bridge(rp(:,cp,ip),rp(:,mcm,kp),std,
     :             rxp,out1)
              if (ip.ne.kp) then
                 call bridge(rp(:,cp,kp),rp(:,mcm,ip),std,
     :                rxpp,out2)
              endif
           else
              call bridge(rp(:,cp,ip),rp(:,mcm,nkp),std,
     :             rxp,out1)
              if (ip.ne.kp) then
                 call bridge(rp(:,cp,kp),rp(:,mcm,nip),std,
     :                rxpp,out2)
              endif
           endif

c     mapping rxp and rxpp from plane to sphere

           vnew=0.d0
           wnew=0.d0
           ll=0
           if(cp+mbm.le.ftn0)then
              do i=cp+1,mcm-1
                 ll=ll+1
                 if (ip.eq.kp) then
                    call ppnergy ( potk, rxp(:,ll), ip,
     :                   rxp(:,ll), kp, i,
     :                   vvv, www )
                 else
                    call ppnergy ( potk, rxp(:,ll), ip,
     :                   rxpp(:,ll), kp, i,
     :                   vvv, www )
                 endif
                 vnew=vnew+vvv
                 wnew=wnew+www
              enddo
           else
              do i=cp+1,ftn0
                 ll=ll+1
                 if (ip.eq.kp) then
                    call ppnergy ( potk, rxp(:,ll), ip,
     :                   rxp(:,ll), kp, i,
     :                   vvv, www )
                 else
                    call ppnergy ( potk, rxp(:,ll), ip,
     :                   rxpp(:,ll), kp, i,
     :                   vvv, www )
                 endif
                 vnew=vnew+vvv
                 wnew=wnew+www
              enddo
              do i=1,mcm-1
                 ll=ll+1
                 if (ip.eq.kp) then
                    call ppnergy ( potk, rxp(:,ll), nip,
     :                   rxp(:,ll), nkp, i,
     :                   vvv, www )
                 else
                    call ppnergy ( potk, rxpp(:,ll), nip,
     :                   rxp(:,ll), nkp, i,
     :                   vvv, www )
                 endif
                 vnew=vnew+vvv
                 wnew=wnew+www
              enddo
           endif
           deltv=vnew-vold

c     execute the bridge move
           acatmab = acatmab + 1.d0
           call updaten(cp,rxp,ip,nkp)
           if (ip.ne.kp) then
              call updaten(cp,rxpp,kp,nip)
              call swapn(cp,ip,nip,kp,nkp)
              call switch(next(ip),next(kp))
              call switch(mext(cp,ip),mext(cp,kp))
              acatmas = acatmas + 1.d0
           endif

c     calculate the ratio of transition probabilities
           call gauss(cp,ga)

c     inverse mapping
           call equiv(rp,rpn)
           call equiv(rpn,rx)
           call mappingi()

           kenew=0.d0
           do k=1,np
              if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
                 do i=1,ftn0
                    call kknergy ( rx(:,i,k), k, i,
     :                   kkk )
                    kenew=kenew+kkk
                 enddo
              endif
           enddo
           deltke=kenew-keold

c     calculate the acceptance probability
           call acc_p(deltv,deltke,ps)
           ps=ps*ga

c     accept/reject
           if (ps.gt.ranf(dummy)) then
              ke=ke+deltke
              v=v+vnew-vold
              w=w+wnew-wold
           else
              call equiv(rx,rpn)
              acatmab = acatmab - 1.d0
              if (ip.ne.kp) then
                 call switch(next(ip),next(kp))
                 call switch(mext(cp,ip),mext(cp,kp))
                 acatmas = acatmas - 1.d0
              endif
           endif

c     build the permutations cycles in lext
           call permcyc()
c     count number of swaps of two particles
           call cycles(cyc,nswap)

c     balance ke between bridge and displace
c          call sumupke ( ke )

           IF (STEP.GT.IEQUI) THEN
              ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

              IF (POTK .EQ. 'LJ') THEN
                 VN = ( V + VLRC )
              ELSE
                 VN = V - W
                 WN = W
              ENDIF
              CALL CWIND( WIND )
              IF (WIND.GT.3*BETA*IC*2/FTM**2) WIND=3*BETA*IC*2/FTM**2

C     ** ACCUMULATE AVERAGES **

              ACV    = ACV    + VN
              ACVSQ  = ACVSQ  + VN*VN
              ACW    = ACW    + WN
              ACWSQ  = ACWSQ  + WN*WN
              ACKE   = ACKE   + KE
              ACKESQ = ACKESQ + KE*KE
              ACWIND = ACWIND + WIND
           ENDIF

           IF (IFBRIDGE) GOTO 97

C     ****************************************************************
C     ** ENDS BRIDGE&SWAP MOVE                                      **
C     ****************************************************************

 77        CONTINUE

C     ****************************************************************
C     ** DISPLACEMENT MOVE                                          **
C     ****************************************************************

C     ** PREVIOUS IP **

           DO I=1,NP
              J=NEXT(I)
              PREV(J)=I
           ENDDO

           NIP = NEXT(IP)
           PIP = PREV(IP)

           DO 90 C1 = 1, FTN0

              DO IDIM = 1, DIM
                 RXIOLD(IDIM) = RX(IDIM,C1,IP)
              ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE OLD CONFIGURATION **

              CALL PENERGY ( POTK, RXIOLD, IP, C1,
     :             VOLD, WOLD )

              CALL KENERGY ( RXIOLD, IP, C1,
     :             KEOLD )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

              ACTOLD = KEOLD + VOLD

C     ** MOVE I AND PICKUP THE CENTRAL IMAGE **

              DO IDIM = 1, DIM
                 RXINEW(IDIM) = RXIOLD(IDIM) +
     :                ( 2.0 * RANF ( DUMMY ) - 1.0 )*DRMAX
                 RXINEW(IDIM) = RXINEW(IDIM) -
     :                DNINT ( RXINEW(IDIM)/SIGM(IDIM) )*SIGM(IDIM)
              ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE NEW CONFIGURATION **
```

```
            CALL PENERGY ( POTK, RXINEW, IP, C1,
     :           VNEW, WNEW )

            CALL KENERGY ( RXINEW, IP, C1,
     :           KENEW )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

            ACTNEW = KENEW + VNEW

            DELTV  = VNEW  - VOLD
            DELTW  = WNEW  - WOLD
            DELTKE = KENEW - KEOLD

C     ** CHECK FOR ACCEPTANCE **

            DELTACT  = ACTNEW - ACTOLD
            DELTACTB = LS*DELTACT

            IF ( EXP ( - DELTACTB ) .GT. RANF ( DUMMY ) ) THEN
               V          = V  + DELTV
               W          = W  + DELTW
               KE         = KE + DELTKE
               ACATMA = ACATMA + 1.0

               DO IDIM = 1, DIM
                  RX(IDIM,C1,IP)  = RXINEW(IDIM)
C     IMAGINARY TIME PERIODIC BOUNDARY CONDITIONS
                  IF (C1.EQ.1) THEN
                     RX(IDIM,FTNO+1,PIP)=RX(IDIM,C1,IP)
                  ENDIF
                  IF (C1.EQ.FTNO) THEN
                     RX(IDIM,0,NIP)=RX(IDIM,C1,IP)
                  ENDIF
               ENDDO
            ENDIF

            IF (STEP.GT.IEQUI) THEN
               ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

C     HISTOGRAM FOR g(r)
               IF ( MOD ( C1, ICALCG ) .EQ. 0 ) THEN
                  CALL GOFR( C1, MAXBIN, HIST )
               ENDIF

               IF (POTK .EQ. 'LJ') THEN
                  VN = ( V + VLRC )
               ELSE
                  VN = V - W
                  WN = W
               ENDIF
               CALL CWIND( WIND )
               IF (WIND.GT.3*BETA*IC*2/FTM**2) WIND=3*BETA*IC*2/FTM**2

C     ** ACCUMULATE AVERAGES **

               ACV    = ACV    + VN
               ACVSQ  = ACVSQ  + VN*VN
               ACW    = ACW    + WN
               ACWSQ  = ACWSQ  + WN*WN
               ACKE   = ACKE   + KE
               ACKESQ = ACKESQ + KE*KE
               ACWIND = ACWIND + WIND
            ENDIF

C     ****************************************************************
C     ** END DISPLACEMENT MOVE                                      **
C     ****************************************************************

 90         CONTINUE

C     ****************************************************************
C     ** ENDS LOOP OVER TIME SLICES                                 **
C     ****************************************************************

 97         CONTINUE

C     ** WRITE OUT THE INSTANTANEOUS VALUES ON FORT.9 **

            SFI = WIND  *FTM**2/BETA/IC/2
            SF  = ACWIND*FTM**2/BETA/IC/2/ACM

            CALL FLUSH(9)
            IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN
               WRITE(9,*) STEP, DIM*NP/2/LS-KE/FTNO, VN/FTNO, WN/FTNO, SFI
            ENDIF

C     ** CREATE POSITION DISTRIBUTION **

C           CALL DISTR(30_8,STEP*FTNO*NP/30)

C     ** PERFORM PERIODIC OPERATIONS **

            IF ( MOD ( STEP, IRATIO ) .EQ. 0 ) THEN

C     ** ADJUST MAXIMUM DISPLACEMENT **

               RATIO  = ACATMA  / DBLE ( FTNO * IRATIO )
               RATIOB = ACATMAB / DBLE ( IRATIO )
               RATIOS = ACATMAS / DBLE ( IRATIO )

               IF ( RATIO .GT. 0.5 ) THEN

C                  DRMAX  = DRMAX  * 1.05

               ELSE

C                  DRMAX  = DRMAX  * 0.95
```

```
               ENDIF

               ACATMA  = 0.0
               ACATMAB = 0.0
               ACATMAS = 0.0

            ENDIF

            IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN

C     ** WRITE OUT RUNTIME INFORMATION **

               WRITE(*,*) '          step            acm  dr/sig    ard
     :     arb       ars'
               WRITE(*,'(2(2XI13),(2Xf6.4),4(2Xf8.6))') STEP, INT(ACM),
     :                                  ALPHA, RATIO, RATIOB, RATIOS
               WRITE(*,*) '            ke              v              w
     :          sf'
               WRITE(*,'(4(2XE13.7))') KE/FTNO, VN/FTNO, WN/FTNO,
     :                                  SFI
               WRITE(*,*) '          <ke>            <v>            <w>
     :         <sf>'
               WRITE(*,'(4(2XE13.7))') ACKE/ACM/FTNO, ACV/ACM/FTNO,
     :                                  ACW/ACM/FTNO, SF
               WRITE(*,*) '         nswap'
               WRITE(*,'(I3)') NSWAP
            ENDIF

            IF ( MOD ( STEP, ISAVE ) .EQ. 0 ) THEN

C     ** WRITE OUT THE CONFIGURATION AT INTERVALS **

               CALL WRITCN ( CNFILE )
               REWIND(16)
               CALL FLUSH(16)
               WRITE(16,*)(HIST(I),I=1,MAXBIN),
     &            ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWIND,ACM,STEP

C          ** WRITE OUT THE g(r) **

               IF ( STEP .GT. IEQUI ) THEN
                  TSTEP = DBLE((STEP - IEQUI)*INT(FTNO/ICALCG))
                  OPEN (UNIT = 10, FILE = 'rdf.dat', STATUS = 'UNKNOWN')
                  DO BIN = 1, MAXBIN
                     RLOWER = DBLE ( BIN - .5)*DELR
                     RUPPER = RLOWER + DELR
                     NIDEAL = (RUPPER/2/RADIUS)**DIM-
     :                    (RLOWER/2/RADIUS)**DIM
                     GR = DBLE(HIST(BIN))/TSTEP/DBLE(NP)**2/NIDEAL
                     CALL FLUSH(10)
                     WRITE (10,*) DBLE(BIN-1)*DELR, GR
                  ENDDO
                  CLOSE (UNIT = 10)
               ENDIF

            ENDIF

 100    CONTINUE

C     ****************************************************************
C     ** ENDS THE LOOP OVER CYCLES                                  **
C     ****************************************************************

         WRITE(*,'(//'' END OF MARKOV CHAIN          ''//)')

C     ** CHECKS FINAL VALUE OF THE POTENTIAL ENERGY IS CONSISTENT **

         CALL SUMUP (POTK, OVRLAP, KEEND, VEND, WEND)

         IF ( ABS(VEND - V) .GT. 1.0d-03 ) THEN

            WRITE(*,'(''  PROBLEM WITH V ENERGY !!!'')')
            WRITE(*,'('' VEND   = '', E20.6)') VEND
            WRITE(*,'('' V      = '', E20.6)') V

         ENDIF

         IF ( ABS(KEEND - KE) .GT. 1.0d-03 ) THEN

            WRITE(*,'(''  PROBLEM WITH KE ENERGY !!!'')')
            WRITE(*,'('' KEEND  = '', E20.6)') KEEND
            WRITE(*,'('' KE     = '', E20.6)') KE

         ENDIF

C     ** WRITE OUT THE FINAL CONFIGURATION FROM THE RUN **

         CALL WRITCN ( CNFILE )

C     ** CALCULATE AND WRITE OUT RUNNING AVERAGES **

         AVV   = ACV / ACM
         ACVSQ = ( ACVSQ / ACM ) - AVV ** 2
         AVKE   = ACKE / ACM
         ACKESQ = ( ACKESQ / ACM ) - AVKE ** 2

C    ** CALCULATE FLUCTUATIONS **

         IF ( ACVSQ  .GT. 0.0 ) FLV  = SQRT ( ACVSQ/ACM )/FTNO
         IF ( ACKESQ .GT. 0.0 ) FLKE = SQRT ( ACKESQ/ACM )/FTNO

         WRITE(*,'(/'' AVERAGES ''/ )')
         WRITE(*,'('' <V/N>                 = '',E12.6)') AVV/FTNO
         WRITE(*,'('' <KE/N>                = '',E12.6)') AVKE/FTNO

         WRITE(*,'(/'' FLUCTUATIONS ''/)')

         WRITE(*,'('' FLUCTUATION IN <V/N>  = '',E12.6)') FLV
         WRITE(*,'('' FLUCTUATION IN <KE/N> = '',E12.6)') FLKE
         WRITE(*,'(/'' END OF SIMULATION ''/)')
```

```
C     ****************************************************************
C     ****************************************************************
C     ****************************************************************
C     ** THE END                                                    **
C     ****************************************************************
C     ****************************************************************
C     ****************************************************************
      CLOSE(UNIT = 16)
      STOP
      END


      subroutine permcyc()
      implicit none
!     given a particle i, lext(i) returns the cycle it belongs to
      INCLUDE       'mc-sph.par'

      integer*8 i,j,k,ii(mnp),jj,kk,ll

      kk=1
      do i=1,np
         ll=1
         ii(ll)=i
 1       jj=next(ii(ll))
         do j=1,ll
            if(jj.eq.ii(j))goto 2
         enddo
         ll=ll+1
         ii(ll)=jj
         goto 1
 2       do k=1,ll
            lext(ii(k))=kk
         enddo
         kk=kk+1
      enddo

      return
      end


      subroutine cycles(cyc,ns)
      implicit none
!     cyc(i) returns the number of particles in cycle of type i
!     ns returns the total number of 2 particles swaps
      INCLUDE       'mc-sph.par'
      integer*8 i,cyc(mnp),ns

      ns=0
      cyc=0
      do i=1,np
         cyc(lext(i))=cyc(lext(i))+1
      enddo
      do i=1,np
         if (cyc(i).ne.0) ns=ns+cyc(i)-1
      enddo

      return
      end


      subroutine switch(i,j)
!     switch i and j
      implicit none
      integer*8 i,j,k
      k=i
      i=j
      j=k
      return
      end


      subroutine acc_p(dv,dke,p)
      implicit none
!     complete acceptance probability
      INCLUDE       'mc-sph.par'

      real*8 p,dke,dv

      p=exp(-ls*dv)        ! contribution from the pair potential
      p=p*exp(-ls*dke)     ! contribution from the kinetic energy

      return
      end


      subroutine updaten(j,rxp,ip,nip)
      implicit none
!     updates a portion of the current path x using the proposed path xp
      INCLUDE       'mc-sph.par'

      integer*8 ip,nip,kp,nkp,j,l,k,idim
      integer*8 mcm
      real*8 rxp(mdim,0:n)

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      l=0
      if(j+mbm.le.ftn0)then
         do k=j+1,mcm-1
            l=l+1
            do idim=1,dim
               rpn(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
      else
         do k=j+1,ftn0
            l=l+1
            do idim=1,dim
               rpn(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
         do k=1,mcm-1
            l=l+1
            do idim=1,dim
               rpn(idim,k,nip)=rxp(idim,l)
            enddo
         enddo
      endif
      do idim=1,dim
         rpn(idim,ftn0+1,ip)=rpn(idim,1,nip)
         rpn(idim,0,nip)=rpn(idim,ftn0,ip)
      enddo

      return
      end


      subroutine swapn(j,ip,nip,kp,nkp)
      implicit none
!     swap the final part of the paths
      INCLUDE       'mc-sph.par'

      integer*8 ip,nip,kp,nkp,j,k,idim
      integer*8 mcm
      real*8    rr

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      if(j+mbm.le.ftn0)then
         do k=mcm,ftn0
            do idim=1,dim
               rr=rpn(idim,k,ip)
               rpn(idim,k,ip)=rpn(idim,k,kp)
               rpn(idim,k,kp)=rr
            enddo
         enddo
         do idim=1,dim
            rpn(idim,ftn0+1,ip)=rpn(idim,1,nkp)
            rpn(idim,ftn0+1,kp)=rpn(idim,1,nip)
            rpn(idim,0,nip)=rpn(idim,ftn0,kp)
            rpn(idim,0,nkp)=rpn(idim,ftn0,ip)
         enddo
      endif
      return
      end


      subroutine bridge(x0,x1,std,xnew,out)
      implicit none
!     sample m gaussians with std from xnew(0)=x0 to xnew(ftn0)=x1
      INCLUDE       'mc-sph.par'

      integer*8 l1,l2,l3,j,idim,out
      real*8 std,d,s,xi
      real*8 x0(mdim),x1(mdim),xnew(mdim,0:n)

      l3=mbm
      do idim=1,dim
         xnew(idim,0)=x0(idim)
         xnew(idim,l3)=x0(idim)+(x1(idim)-x0(idim))-
     :          dnint((x1(idim)-x0(idim))/sigm(idim))*sigm(idim)
      enddo
      do j=1,mbm-1
         l1=j-1
         l2=j
         s=std*(dble(l3-l2)/dble(l3-l1))**0.5d0
         do idim=1,dim
            d=xnew(idim,l3)-xnew(idim,l1)
            d=d-dnint(d/sigm(idim))*sigm(idim)
            xnew(idim,j)=xnew(idim,l1)+d/dble(l3-l1)+xi(s)
            xnew(idim,j)=xnew(idim,j)-
     :              dnint(xnew(idim,j)/sigm(idim))*sigm(idim)
         enddo
      enddo
      return
      end


      function xi(std)
!     sample a gaussian with standard deviation std (box-muller method)
      implicit none
      real*8 xi,std,pi,ranf
      data pi/3.141592653589793238462643383 28d0/
      xi=cos(pi*ranf(0.d0))*std*sqrt(-2.d0*log(tiny(pi)+ranf(0.d0)))
      return
      end


      SUBROUTINE DISTR(NN,NORM)
      IMPLICIT NONE
C     WRITES ON FORT.10 THE X-POSITION DISTRIBUTION
      INCLUDE       'mc-sph.par'

      REAL*8        DS,DIST(0:1000)
      INTEGER*8     NN,NORM,I,J,K
      SAVE          DIST

      DS=SIGMA/NN

      DO I=0,NN
         DO J=1,FTN0
            DO K=1,NP
               IF(-SIGMA/2+(I-.5)*DS.LT.RX(1,J,K).AND.
     :              RX(1,J,K).LT.-SIGMA/2+(I+.5)*DS) THEN
                  DIST(I)=DIST(I)+1.D0
               ENDIF
            ENDDO
         ENDDO
         WRITE(10,*) -SIGMA/2+I*DS,DIST(I)/NORM
      ENDDO

      CLOSE(UNIT=10)
```

```
      RETURN
      END


      FUNCTION FACT ( N )
      IMPLICIT NONE
C     FACTORIAL FUNCTION
      INTEGER*8 FACT,N,P,I
      P=1
      DO I=1,N
         P=P*I
      ENDDO
      FACT=P
      END


      SUBROUTINE SUMUP (POTK, OVRLAP, KE, V, W)
      IMPLICIT NONE
C    ****************************************************************
C    ** CALCULATES THE TOTAL ENERGY                                **
C    **                                                            **
C    ** USAGE:                                                     **
C    **                                                            **
C    ** THE SUBROUTINE RETURNS THE TOTAL ENERGY AT THE             **
C    ** BEGINNING AND END OF THE RUN.                              **
C    ****************************************************************
      INCLUDE       'mc-sph.par'

      REAL*8        V, KE, VV, KK
      LOGICAL       OVRLAP
      CHARACTER     POTK*(*)

      REAL*8        RXII, RXIJ, RYIJ, RZIJ
      REAL*8        VIJ, WIJ, RIJSQ, W, WW

      INTEGER*8     TAU, I, J, IDIM

C     POTENTIAL ACTION

      VV      = 0.0
      WW      = 0.0

C    ** LOOP OVER ALL THE PAIRS IN THE LIQUID **

      DO TAU = 1, FTNO
         DO 100 I = 1, NP - 1
            DO 99 J = I + 1, NP
C               RIJSQ = 0.d0
C               DO IDIM = 1, DIM
C                  RXIJ = RX(IDIM,TAU,I) - RX(IDIM,TAU,J)
C     ** MINIMUM IMAGE THE PAIR SEPARATIONS **
C                  RXIJ = RXIJ -
C     :                 DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                  RIJSQ = RIJSQ + RXIJ * RXIJ
C               ENDDO

C     ** EUCLIDEAN DISTANCE
               RXIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
     :              COS(RX(2,TAU,I))*COS(RX(2,TAU,J))
               RYIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
     :              SIN(RX(2,TAU,I))*SIN(RX(2,TAU,J))
               RZIJ  = SIN(RX(1,TAU,I))*SIN(RX(1,TAU,J))
               RIJSQ = RXIJ + RYIJ + RZIJ
               RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

               CALL POT (RIJSQ, VIJ, WIJ, POTK)
               VV   = VV + VIJ
C               WW   = WW + WIJ
 99          CONTINUE
             VV = VV - LOG(ABS(RADIUS**2*COS(RX(1,TAU,I))))/LS
             WW = WW - LOG(ABS(RADIUS**2*COS(RX(1,TAU,I))))/LS
 100      CONTINUE
          VV = VV - LOG(ABS(RADIUS**2*COS(RX(1,TAU,NP))))/LS
          WW = WW - LOG(ABS(RADIUS**2*COS(RX(1,TAU,NP))))/LS
      ENDDO
      V = VV
      W = WW


C     KINETIC ACTION

      KK     = 0.0

      DO TAU = 1, FTNO
         DO I = 1, NP
            RXII = RX(1,TAU,I)-RX(1,TAU+1,I)
            RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
            KK=KK+FTM*(RADIUS*RXII)**2./(2.D0*LS**2.)

            RXII = RX(2,TAU,I)-RX(2,TAU+1,I)
            RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
            KK=KK+FTM*(RADIUS*COS(RX(1,TAU,I))*RXII)**2./(2.D0*LS**2.)
         ENDDO
      ENDDO
      KE = KK

      RETURN
      END


      SUBROUTINE SUMUPKE ( KE )
      IMPLICIT NONE
C    ****************************************************************
C    ** CALCULATES THE TOTAL ENERGY                                **
C    **                                                            **
C    ** USAGE:                                                     **
C    **                                                            **
C    ** THE SUBROUTINE RETURNS THE TOTAL ENERGY AT THE             **
C    ** BEGINNING AND END OF THE RUN.                              **
C    ****************************************************************
      INCLUDE       'mc-sph.par'

      REAL*8        KE, KK, RXII
      INTEGER*8     TAU, I

C     KINETIC ACTION

      KK     = 0.0

      DO TAU = 1, FTNO
         DO I = 1, NP
            RXII = RX(1,TAU,I)-RX(1,TAU+1,I)
            RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
            KK=KK+FTM*(RADIUS*RXII)**2./(2.D0*LS**2.)

            RXII = RX(2,TAU,I)-RX(2,TAU+1,I)
            RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
            KK=KK+FTM*(RADIUS*COS(RX(1,TAU,I))*RXII)**2./(2.D0*LS**2.)
         ENDDO
      ENDDO
      KE = KK

      RETURN
      END


      SUBROUTINE PENERGY ( POTK, RXI, I, TAU,
     :    V, W )
      IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOM I WITH ALL OTHER ATOMS.  **
C    **                                                            **
C    ** PRINCIPAL VARIABLES:                                       **
C    **                                                            **
C    ** INTEGER I                 THE ATOM OF INTEREST             **
C    ** INTEGER NP                THE NUMBER OF ATOMS              **
C    ** INTEGER TAU               THE TIMESTEP                     **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS             **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I      **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I           **
C    **                                                            **
C    ** USAGE:                                                     **
C    **                                                            **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY  **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND     **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                        **
C    ****************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        RXI(MDIM), V, W
      INTEGER*8     I, J, TAU, IDIM
      CHARACTER     POTK*(*)

      REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, VIJ, WIJ

      V      = 0.0
      W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I  **

      DO J = 1, NP

         IF ( I .NE. J ) THEN
            RIJSQ = 0.d0
C             DO IDIM = 1, DIM
C                RXIJ = RXI(IDIM) - RX(IDIM,TAU,J)

C                RXIJ = RXIJ -
C     :               DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                RIJSQ = RIJSQ + RXIJ * RXIJ
C             ENDDO

            RXIJ  = COS(RXI(1))*COS(RX(1,TAU,J))*
     :             COS(RXI(2))*COS(RX(2,TAU,J))
            RYIJ  = COS(RXI(1))*COS(RX(1,TAU,J))*
     :             SIN(RXI(2))*SIN(RX(2,TAU,J))
            RZIJ  = SIN(RXI(1))*SIN(RX(1,TAU,J))
            RIJSQ = RXIJ + RYIJ + RZIJ
            RIJSQ = RADIUS**2*(2*(1.-RIJSQ))
            CALL POT (RIJSQ, VIJ, WIJ, POTK)
            V   = V + VIJ
C            W   = W + WIJ
         ENDIF

      ENDDO

      V = V - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS
      W = W - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS

      RETURN
      END


      SUBROUTINE PPPNERGY ( POTK, RXI, I, RXJ, J,
     :    TAU, V, W )
      IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOMS I AND J WITH         **
C    ** ALL OTHER ATOMS PLUS THE ONE BETWEEN I AND J               **
C    **                                                            **
C    ** PRINCIPAL VARIABLES:                                       **
C    **                                                            **
C    ** INTEGER I, J              THE ATOMS OF INTEREST            **
C    ** INTEGER NP                THE NUMBER OF ATOMS              **
C    ** INTEGER TAU               THE TIMESTEP                     **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS             **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I      **
C    ** REAL*8    RXJ,RYJ,RZJ       THE COORDINATES OF ATOM J      **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I           **
```

```
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                           **
C    ******************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        RXI(MDIM), RXJ(MDIM), V, W
      INTEGER*8     I, J, K, TAU, IDIM
      CHARACTER     POTK*(*)

      REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, VIJ, WIJ

      V      = 0.0
      W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I AND J **

      DO 100 K = 1, NP
         IF ( K .NE. I .AND. K .NE. J ) THEN
            RIJSQ = 0.d0
C            DO IDIM = 1, DIM
C               RXIJ = RXI(IDIM) - RX(IDIM,TAU,K)

C               RXIJ = RXIJ -
C     :               DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C               RIJSQ = RIJSQ + RXIJ * RXIJ
C            ENDDO

C     ** EUCLIDEAN DISTANCE
            RXIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :            COS(RXI(2))*COS(RX(2,TAU,K))
            RYIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :            SIN(RXI(2))*SIN(RX(2,TAU,K))
            RZIJ  = SIN(RXI(1))*SIN(RX(1,TAU,K))
            RIJSQ = RXIJ + RYIJ + RZIJ
            RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

            CALL POT (RIJSQ, VIJ, WIJ, POTK)
            V   = V + VIJ
C            W   = W + WIJ

            IF ( I .NE. J ) THEN
               RIJSQ = 0.d0
C               DO IDIM = 1, DIM
C                  RXIJ = RXJ(IDIM) - RX(IDIM,TAU,K)

C                  RXIJ = RXIJ -
C     :                DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                  RIJSQ = RIJSQ + RXIJ * RXIJ
C               ENDDO

C     ** EUCLIDEAN DISTANCE
               RXIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :               COS(RXJ(2))*COS(RX(2,TAU,K))
               RYIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :               SIN(RXJ(2))*SIN(RX(2,TAU,K))
               RZIJ  = SIN(RXJ(1))*SIN(RX(1,TAU,K))
               RIJSQ = RXIJ + RYIJ + RZIJ
               RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

               CALL POT (RIJSQ, VIJ, WIJ, POTK)
               V   = V + VIJ
C               W   = W + WIJ

            ENDIF

         ENDIF

 100    CONTINUE

C        RETURN

      V = V - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS
      W = W - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS

      IF (I .NE. J) THEN
         RIJSQ = 0.d0
C          DO IDIM = 1, DIM
C             RXIJ = RXI(IDIM) - RXJ(IDIM)

C             RXIJ = RXIJ -
C     :            DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C             RIJSQ = RIJSQ + RXIJ * RXIJ
C          ENDDO

C     ** EUCLIDEAN DISTANCE
         RXIJ  = COS(RXI(1))*COS(RXJ(1))*
     :          COS(RXI(2))*COS(RXJ(2))
         RYIJ  = COS(RXI(1))*COS(RXJ(1))*
     :          SIN(RXI(2))*SIN(RXJ(2))
         RZIJ  = SIN(RXI(1))*SIN(RXJ(1))
         RIJSQ = RXIJ + RYIJ + RZIJ
         RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

         CALL POT (RIJSQ, VIJ, WIJ, POTK)
         V   = V + VIJ
C          W   = W + WIJ
         V = V - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
         W = W - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
      ENDIF

      RETURN
      END


      SUBROUTINE PPNERGY ( POTK, RPI, I, RPJ, J,
     :    TAU, V, W )
      IMPLICIT NONE
```

```
C    ******************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOMS I AND J WITH            **
C    ** ALL OTHER ATOMS PLUS THE ONE BETWEEN I AND J                  **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                          **
C    **                                                              **
C    ** INTEGER I, J              THE ATOMS OF INTEREST               **
C    ** INTEGER NP                THE NUMBER OF ATOMS                 **
C    ** INTEGER TAU               THE TIMESTEP                        **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS                **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I         **
C    ** REAL*8    RXJ,RYJ,RZJ       THE COORDINATES OF ATOM J         **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I    **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I              **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                           **
C    ******************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        RPI(MDIM),RPJ(MDIM),RXI(MDIM),RXJ(MDIM), V, W
      INTEGER*8     I, J, K, TAU, IDIM
      CHARACTER     POTK*(*)

      REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, VIJ, WIJ

C     MAPPING
      RXI(1)=RPI(1)
      RXI(2)=RPI(2)
      RXJ(1)=RPJ(1)
      RXJ(2)=RPJ(2)

      V      = 0.0
      W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I AND J **

      DO 100 K = 1, NP
         IF ( K .NE. I .AND. K .NE. J ) THEN
            RIJSQ = 0.d0
C            DO IDIM = 1, DIM
C               RXIJ = RXI(IDIM) - RX(IDIM,TAU,K)

C               RXIJ = RXIJ -
C     :               DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C               RIJSQ = RIJSQ + RXIJ * RXIJ
C            ENDDO

C     ** EUCLIDEAN DISTANCE
            RXIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :            COS(RXI(2))*COS(RX(2,TAU,K))
            RYIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :            SIN(RXI(2))*SIN(RX(2,TAU,K))
            RZIJ  = SIN(RXI(1))*SIN(RX(1,TAU,K))
            RIJSQ = RXIJ + RYIJ + RZIJ
            RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

            CALL POT (RIJSQ, VIJ, WIJ, POTK)
            V   = V + VIJ
C            W   = W + WIJ

            IF ( I .NE. J ) THEN
               RIJSQ = 0.d0
C               DO IDIM = 1, DIM
C                  RXIJ = RXJ(IDIM) - RX(IDIM,TAU,K)

C                  RXIJ = RXIJ -
C     :                DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                  RIJSQ = RIJSQ + RXIJ * RXIJ
C               ENDDO

C     ** EUCLIDEAN DISTANCE
               RXIJ  = COS(RXJ(1))*COS(RX(1,TAU,K))*
     :               COS(RXJ(2))*COS(RX(2,TAU,K))
               RYIJ  = COS(RXJ(1))*COS(RX(1,TAU,K))*
     :               SIN(RXJ(2))*SIN(RX(2,TAU,K))
               RZIJ  = SIN(RXJ(1))*SIN(RX(1,TAU,K))
               RIJSQ = RXIJ + RYIJ + RZIJ
               RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

               CALL POT (RIJSQ, VIJ, WIJ, POTK)
               V   = V + VIJ
C               W   = W + WIJ

            ENDIF

         ENDIF

 100    CONTINUE

C        RETURN

      V = V - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS
      W = W - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS

      IF (I .NE. J) THEN
         RIJSQ = 0.d0
C          DO IDIM = 1, DIM
C             RXIJ = RXI(IDIM) - RXJ(IDIM)

C             RXIJ = RXIJ -
C     :            DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C             RIJSQ = RIJSQ + RXIJ * RXIJ
C          ENDDO

C     ** EUCLIDEAN DISTANCE
         RXIJ  = COS(RXI(1))*COS(RXJ(1))*
     :          COS(RXI(2))*COS(RXJ(2))
```

```
         RYIJ  = COS(RXI(1))*COS(RXJ(1))*
     :         SIN(RXI(2))*SIN(RXJ(2))
         RZIJ  = SIN(RXI(1))*SIN(RXJ(1))
         RIJSQ = RXIJ + RYIJ + RZIJ
         RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

         CALL POT (RIJSQ, VIJ, WIJ, POTK)
         V   = V + VIJ
C         W   = W + WIJ
         V = V - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
         W = W - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
      ENDIF

      RETURN
      END


      SUBROUTINE KENERGY ( RXI, I, TAU, KE )
      IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C     **                                                              **
C     ** PRINCIPAL VARIABLES:                                         **
C     **                                                              **
C     ** INTEGER I                THE ATOM OF INTEREST                **
C     ** INTEGER NP               THE NUMBER OF ATOMS                 **
C     ** INTEGER TAU              THE TIMESTEP                        **
C     ** REAL*8    RX, RY, RZ       THE ATOM POSITIONS                **
C     ** REAL*8    RXI,RYI,RZI      THE COORDINATES OF ATOM I         **
C     ** REAL*8    KE               THE KINETIC ENERGY OF ATOM I      **
C     **                                                              **
C     ** USAGE:                                                       **
C     **                                                              **
C     ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C     ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C     ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        RXI(MDIM), KE
      REAL*8        RXIP1, RXIS1, RXIP2, RXIS2, RXII
      INTEGER*8     I, TAU, IDIM

      KE = 0.d0

      RXIP1 = RX(1,TAU-1,I)
      RXIS1 = RX(1,TAU+1,I)
      RXII = RXI(1) - RXIP1
      RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
      KE=KE+0.5*FTM*(RADIUS*RXII/LS)**2.
      RXII = RXI(1) - RXIS1
      RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
      KE=KE+0.5*FTM*(RADIUS*RXII/LS)**2.

      RXIP2 = RX(2,TAU-1,I)
      RXIS2 = RX(2,TAU+1,I)
      RXII = RXI(2) - RXIP2
      RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
      KE=KE+0.5*FTM*(RADIUS*COS(RXIP1)*RXII/LS)**2.
      RXII = RXI(2) - RXIS2
      RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
      KE=KE+0.5*FTM*(RADIUS*COS(RXI(1))*RXII/LS)**2.

      RETURN
      END


      SUBROUTINE KKNERGY ( RXI, IP, TAU, KE )
      IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C     **                                                              **
C     ** PRINCIPAL VARIABLES:                                         **
C     **                                                              **
C     ** INTEGER I                THE ATOM OF INTEREST                **
C     ** INTEGER NP               THE NUMBER OF ATOMS                 **
C     ** INTEGER TAU              THE TIMESTEP                        **
C     ** REAL*8    RX, RY, RZ       THE ATOM POSITIONS                **
C     ** REAL*8    RXI,RYI,RZI      THE COORDINATES OF ATOM I         **
C     ** REAL*8    KE               THE KINETIC ENERGY OF ATOM I      **
C     **                                                              **
C     ** USAGE:                                                       **
C     **                                                              **
C     ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C     ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C     ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        RXI(MDIM), KE
      REAL*8        RXIS, RXII
      INTEGER*8     IP, TAU, IDIM

      KE = 0.d0

      RXIS = RX(1,TAU+1,IP)
      RXII = RXI(1) - RXIS
      RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
      KE=KE+0.5*FTM*(RADIUS*RXII/LS)**2.

      RXIS = RX(2,TAU+1,IP)
      RXII = RXI(2) - RXIS
      RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
      KE=KE+0.5*FTM*(RADIUS*COS(RXI(1))*RXII/LS)**2.

      RETURN
      END


      SUBROUTINE CWIND ( WIND )
      IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS THE WINDING NUMBER.                                  **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        WIND, WI, SURF
      REAL*8        RX11, RX21, RX31, RX12, RX22, RX32
      INTEGER*8     I, J, IDIM

      WIND = 0.d0
      SURF = 4*PI*RADIUS**2
      DO I = 1, NP
         DO J = 1, FTNO - 1
            RX11 = RX(1,J  ,I)
            RX21 = RX(1,J+1,I)
            RX31 = RX(1,J+2,I)
            RX12 = RX(2,J  ,I)
            RX22 = RX(2,J+1,I)
            RX32 = RX(2,J+2,I)
            WI = ((RX21-RX11)*(RX32-RX22)-(RX22-RX12)*(RX31-RX21))*
     :             RADIUS**2*COS(RX21)
            WIND = WIND + WI
         ENDDO
      ENDDO
      IF (WIND.GT.SURF) WIND = SURF
      WIND = WIND**2

 20   CONTINUE
      RETURN
      END


      REAL*8 FUNCTION RANF ( DUMMY )
      IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS A UNIFORM RANDOM VARIATE IN THE RANGE 0 TO 1.         **
C     **                                                              **
C     **                 ***************                              **
C     **                 **  WARNING  **                              **
C     **                 ***************                              **
C     **                                                              **
C     ** GOOD RANDOM NUMBER GENERATORS ARE MACHINE SPECIFIC.           **
C     ** PLEASE USE THE ONE RECOMMENDED FOR YOUR MACHINE.              **
C     **                                                              **
C     ** RAND(FLAG) returns a pseudo-random number from a uniform      **
C     ** distribution between 0 and 1. If FLAG is 0, the next number   **
C     ** in the current sequence is returned; if FLAG is 1, the        **
C     ** generator is restarted by CALL SRAND(0); if FLAG has any      **
C     ** other value, it is used as a new seed with SRAND.             **
C     ****************************************************************
      INTEGER*8   L, C, M
      PARAMETER ( L = 1029, C = 221591, M = 1048576 )
      INTEGER*8   SEED
      SAVE        SEED
      DATA        SEED / 0 /
      REAL*8      DUMMY

C      SEED = MOD ( SEED * L + C, M )
C      RANF = DBLE( SEED ) / M

      RANF = RAND ( )

      RETURN
      END


      SUBROUTINE READCN ( CNFILE )
      IMPLICIT NONE
C     ****************************************************************
C     ** SUBROUTINE TO READ IN THE CONFIGURATION FROM UNIT 10          **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      CHARACTER   CNFILE*(*), HASH*1, MYFMT*77

      INTEGER*8   CNUNIT, I, J, NNP, NI, IDIM
      PARAMETER ( CNUNIT = 10 )

C    ****************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'OLD' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X))'

      READ ( CNUNIT,* ) HASH, FTNO, NNP
      IF ( NNP .NE. NP ) STOP 'N ERROR IN READCN'

      DO 100 I = 1, NNP
         READ ( CNUNIT,* ) HASH, NEXT(I)
         NI = NEXT(I)
         DO 90 J = 1, FTNO+1
            READ ( CNUNIT, MYFMT) LEXT(I), (RX(IDIM,J,I),IDIM=1,DIM)
 90      ENDDO
         READ ( CNUNIT,13 )
         READ ( CNUNIT,13 )
         DO IDIM = 1, DIM
            RX(IDIM,0,NI) = RX(IDIM,FTNO,I)
         ENDDO
 100  ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE WRITCN ( CNFILE )
      IMPLICIT NONE
C     ****************************************************************
C     ** SUBROUTINE TO WRITE OUT THE CONFIGURATION TO UNIT 10          **
C     ****************************************************************
```

```
      INCLUDE     'mc-sph.par'

      CHARACTER   CNFILE*(*), MYFMT*77

      INTEGER*8   CNUNIT, I, J, IDIM
      REAL*8      R
      PARAMETER ( CNUNIT = 10 )

C   ******************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X))'

      WRITE(*,*) 'output to file -----------------------------------
     :---------------'
      CALL FLUSH ( CNUNIT )
      WRITE ( CNUNIT,* ) '#',FTNO,NP

      DO I = 1, NP
         WRITE ( CNUNIT,* ) '#', NEXT(I)
         DO J = 1, FTNO+1
C            WRITE ( CNUNIT, MYFMT ) LEXT(I),
C     :    (RX(IDIM,J,I)-DNINT(RX(IDIM,J,I)/SIGMA)*SIGMA,IDIM=1,DIM)
            WRITE ( CNUNIT, MYFMT ) LEXT(I),
     :            (RX(IDIM,J,I),IDIM=1,DIM)
         ENDDO
         WRITE ( CNUNIT,13 ) ''
         WRITE ( CNUNIT,13 ) ''
      ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE INITCNSPH ( CNFILE )
      IMPLICIT NONE
C    ******************************************************************
C    ** SUBROUTINE TO INITIALIZE THE CONFIGURATION TO UNIT 10         **
C    ******************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8      RANF
      CHARACTER   CNFILE*(*), MYFMT*77

      INTEGER*8   CNUNIT, I, J, IDIM
      REAL*8      RXI(DIM)
      PARAMETER ( CNUNIT = 10 )

C   ******************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X))'

      WRITE ( CNUNIT,* ) '#',FTNO,NP

      DO I = 1, NP
         WRITE ( CNUNIT,* ) '#', NEXT(I)
         RXI(1) = 0.
         RXI(2) = SIGM(2)*(RANF(0.d0)-.5)
         DO J = 1, FTNO+1
            WRITE ( CNUNIT, MYFMT ) I, RXI(1), RXI(2)
         ENDDO
         WRITE ( CNUNIT,13 ) ''
         WRITE ( CNUNIT,13 ) ''
      ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE POT (RIJSQ, VIJ, WIJ, POTK)
      IMPLICIT NONE
C    ******************************************************************
C    ** SUBROUTINE FOR THE PAIR POTENTIAL                            **
C    ******************************************************************
      INCLUDE     'mc-sph.par'

      CHARACTER POTK*(*)
      REAL*8    RIJ, RIJSQ, RMIN, RMSQ, RCSQ, SR2, SR6, VIJ, WIJ
      REAL*8    COU3, QHS3, F
      PARAMETER ( COU3 = 4.760154727959107013287634700 57d0 )
      PARAMETER ( QHS3 = 24.d0 )

C       PI = ACOS(-1.d0)

      VIJ = 0.d0
      WIJ = 0.d0
      IF (DIM .GT. 3) THEN
         WRITE(*,*) "DIM > 3 !!!"
         STOP
      ENDIF
      IF (POTK .EQ. 'FREE') RETURN

      IF (POTK .EQ. 'LJ') THEN
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C     W. L. McMillan, Phys. Rev. A 138, 442 (1965)     C
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C          RMIN = 2.d0**(1./6)
C          RMIN = EPSA
C          RMIN = TINY(PI)
         RMIN = 0.d0
         RIJSQ = RIJSQ / (SIG * SIG)
         RMSQ = RMIN*RMIN
         RCSQ = RCUT*RCUT
         IF (RIJSQ .LT. RMSQ) THEN
            RIJSQ = RMSQ
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ELSEIF (RIJSQ .GT. RMSQ .AND. RIJSQ .LT. RCSQ) THEN
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ENDIF
      ELSEIF (POTK .EQ. 'BUMP') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG) THEN
            VIJ = EPSR
         ENDIF
      ELSEIF (POTK .EQ. 'PSW') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG) THEN
            VIJ = EPSR
         ELSEIF (RIJ .LE. SIG + RCUT .AND. RIJ .GT. SIG) THEN
            VIJ = -EPSA
         ENDIF
      ELSEIF (POTK .EQ. 'COULOMB') THEN
         F = NP/(NP-1)
         RIJ = SQRT( RIJSQ )
         IF (RIJ .GT. TINY(PI)) THEN
            IF (DIM .EQ. 3) THEN
               VIJ = -F*COU3*SIG/SIGMA/2.
               VIJ = VIJ + SIG/RIJ
            ELSEIF (DIM .EQ. 2) THEN
               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/SIG))/4.
               VIJ = VIJ - LOG(RIJ/SIG)
            ELSEIF (DIM .EQ. 1) THEN
               VIJ = F*SIGMA/SIG/4.
               VIJ = VIJ - RIJ/SIG
            ENDIF
         ELSE
            RIJ = TINY(PI)
            IF (DIM .EQ. 3) THEN
               VIJ = -F*COU3*SIG/SIGMA/2.
               VIJ = VIJ + SIG/RIJ
            ELSEIF (DIM .EQ. 2) THEN
               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/SIG))/4.
               VIJ = VIJ - LOG(RIJ/SIG)
            ELSEIF (DIM .EQ. 1) THEN
               VIJ = F*SIGMA/SIG/4.
               VIJ = VIJ - RIJ/SIG
            ENDIF
         ENDIF
      ELSEIF (POTK .EQ. 'QHS') THEN
         F = NP/(NP-1)
         IF (DIM .EQ. 3) THEN
            IF (RIJSQ .LE. SIG*SIG) THEN
               VIJ = 10.**10.
            ELSE
            VIJ = -F*QHS3*(SIGMA-SIG)/SIGMA**3/2./FTM
            VIJ = VIJ + (2.*RIJSQ+SIG**2.)/(RIJSQ-SIG*SIG)**2./2./FTM
            WIJ = -2.*RIJSQ*(RIJSQ+2.*SIG*SIG)/(RIJSQ-SIG*SIG)**3./FTM
            ENDIF
         ELSE
            WRITE(*,*) 'QHS in dimension different from 3'
            STOP
         ENDIF
      ELSEIF (POTK .EQ. 'HARMONIC') THEN
         VIJ = 0.5*RIJSQ/SIG**2.
      ENDIF

      RETURN
      END


      SUBROUTINE GOFR ( TAU, MAXBIN, HIST )
      IMPLICIT NONE
C    ******************************************************************
C    ** CALCULATES THE RADIAL DISTRIBUTION FUNCTION HISTOGRAM.       **
C    ******************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, RIJ
      INTEGER*8     TAU, IDIM, I, J, BIN, MAXBIN, HIST(MAXBIN)

      DO I = 1, NP - 1
         DO J = I + 1, NP
C     CALCULATE HISTOGRAM FOR g(r)

C     Euclidean Distance
            RXIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
     :            COS(RX(2,TAU,I))*COS(RX(2,TAU,J))
            RYIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
     :            SIN(RX(2,TAU,I))*SIN(RX(2,TAU,J))
            RZIJ  = SIN(RX(1,TAU,I))*SIN(RX(1,TAU,J))
            RIJSQ = RXIJ + RYIJ + RZIJ
c            RIJ = 2*RADIUS*SIN(ACOS(RIJSQ)/2.)
            RIJ = RADIUS*SQRT(2.-2*RIJSQ)

C     Geodesic Distance
c            RIJ = RADIUS*ACOS(SIN(RX(1,TAU,I))*SIN(RX(1,TAU,J))+
c     : COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*COS(RX(2,TAU,I)-RX(2,TAU,J)))

            BIN = INT ( RIJ/DELR ) + 1

            IF (BIN .LE. MAXBIN) THEN
               HIST(BIN) = HIST(BIN) + 2
            ENDIF

         ENDDO
```

```
      ENDDO

      RETURN
      END


      SUBROUTINE MAPPINGD()
      IMPLICIT NONE
C     ****************************************************************
C     ** CHANGE COORDINATES (DIRECT)                                **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      INTEGER*8 I,J

      DO I=1,NP
         DO J=0,FTNO+1
            RP(1,J,I)=RX(1,J,I)
            RP(2,J,I)=RX(2,J,I)
         ENDDO
      ENDDO

      RETURN
      END


      SUBROUTINE MAPPINGI()
      IMPLICIT NONE
C     ****************************************************************
C     ** CHANGE COORDINATES (INVERSE)                               **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      INTEGER*8 I,J

      DO I=1,NP
         DO J=0,FTNO+1
            RX(1,J,I)=RP(1,J,I)
            RX(2,J,I)=RP(2,J,I)
         ENDDO
      ENDDO

      RETURN
      END


      SUBROUTINE GAUSS(CP,GA)
      IMPLICIT NONE
C     ****************************************************************
C     ** GAUSSIAN TRANSITION PROBABILITY FOR BROWNIAN BRIDGE        **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      INTEGER*8 I,J,K,CP,MCM
      REAL*8 GA,VAR,DD

      MCM = CP+MBM-FLOOR((CP+MBM-.1)/FTNO)*FTNO
      VAR = 2*LS/FTM

      GA = 0.D0
      DO I=1,DIM
         DO J=1,NP
            IF(CP+MBM.LE.FTNO)THEN
               DO K=CP+1,MCM
                  DD=RPN(I,K,J)-RPN(I,K-1,J)
                  DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                  GA=GA+DD**2
                  DD=RP(I,K,J)-RP(I,K-1,J)
                  DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                  GA=GA-DD**2
               ENDDO
            ELSE
               DO K=CP+1,FTNO
                  DD=RPN(I,K,J)-RPN(I,K-1,J)
                  DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                  GA=GA+DD**2
                  DD=RP(I,K,J)-RP(I,K-1,J)
                  DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                  GA=GA-DD**2
               ENDDO
               DO K=1,MCM
                  DD=RPN(I,K,J)-RPN(I,K-1,J)
                  DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                  GA=GA+DD**2
                  DD=RP(I,K,J)-RP(I,K-1,J)
                  DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                  GA=GA-DD**2
               ENDDO
            ENDIF
         ENDDO
      ENDDO

      GA = EXP(GA/VAR)

      RETURN
      END


      SUBROUTINE EQUIV(A,B)
      IMPLICIT NONE
C     ****************************************************************
C     ** A = B WITH A, B TWO PATHS                                  **
C     ****************************************************************
      INCLUDE     'mc-sph.par'

      INTEGER*8 I,J,K
      REAL*8    A(MDIM,0:N,MNP),B(MDIM,0:N,MNP)

      DO I=1,DIM
         DO J=0,FTNO+1
            DO K=1,NP
               A(I,J,K)=B(I,J,K)
            ENDDO
         ENDDO
      ENDDO

      RETURN
      END
```

```
*****************************************************************************
*** mc-sph.par ***********************************************************
*****************************************************************************

      REAL*8        PI
      REAL*8        DELR
         INTEGER*8     MDIM, N, MNP

         PARAMETER ( PI = 3.14159265358979323846264338328d0 )

      PARAMETER ( MDIM = 10   )   ! maximum number of dimensions
      PARAMETER ( MNP  = 1000 )   ! maximum number of particles
      PARAMETER ( N    = 3000 )   ! maximum number of time slices
      PARAMETER ( DELR = 0.01d0 ) ! grid spacing for g(r)

      INTEGER*8     DIM
         REAL*8        RX(MDIM,0:N,MNP)
      REAL*8        RPN(MDIM,0:N,MNP),RP(MDIM,0:N,MNP)
      REAL*8        FTM, LS, SIGMA, SIGM(MDIM), RADIUS
         REAL*8        SIG, EPSA, EPSR, EPS, RCUT
         INTEGER*8     FTNO, NP
      INTEGER*8     NEXT(MNP), LEXT(MNP), MEXT(N,MNP), MBM

         ! path
         COMMON / BLOCK1 / RX, RP, RPN, DIM
         ! pair-potential parameters
         COMMON / BLOCK2 / SIG, EPSR, EPSA, EPS, RCUT
         ! mass, timestep, box edge, # timeslices, # particles, radius
         COMMON / BLOCK3 / FTM, LS, SIGMA, SIGM, FTNO, NP, RADIUS
         ! permutations
         COMMON / BLOCK4 / NEXT, LEXT, MEXT, MBM

*****************************************************************************
*** data-gr-rs.in ********************************************************
*****************************************************************************

0 number of spatial dimensions          (2 sphere)
2
1 seed of the random sequence                 RAND
3
2 if bose                                    (T/F)
T
3 number of particles                     (<= MNP)
10
4 the potential                     SUBROUTINE POT
FREE
5 # time slices = 1/temperature/timestep     (< N)
50
6 mass                              (hbar = kb = 1)
1.d0
7 # of cycles                               (nstep)
500000000000000000
8 # of steps between output lines         (iprint)
100
9 # of steps between configuration saves   (isave)
100
10 # of steps for equilibration            (iequi)
100
11 # of steps for acceptance ratios       (iratio)
100
12 # of steps for rdf calculation    (icalcg < #5)
10
13 configuration file name
conf.xyz
14 0 init zero, 1 init, 2 rest conf, 3 rest full
1
15 radius of the sphere
5.d0
16 temperature                      THERMODYNAMICS
.03d0
17 maximum displacement/box edge
.05d0
18 maximum # of bridge timeslices     (> 1; <= #5)
10
19 potential cutoff distance (LJ)   SUBROUTINE POT
2.d0
20 sig    (LJ 2.566)                SUBROUTINE POT
1.d0
21 epsr   (LJ INIT)                 SUBROUTINE POT
1.d10
22 epsa                             SUBROUTINE POT
1.d0
23 eps    (LJ 10.22)                SUBROUTINE POT
1.d1
24 if only displace                          (T/F)
F
25 if only bridge                            (T/F)
F
```

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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